week5_6_VelazquezAdrian.R

2026-02-01

```r
## Week5-6_Assignment
## Velazquez-Feliciano, Adrian
## DSC 640: Data Presentation and Visualization
## 02/01/2026

library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.5.2

library(dplyr)

## Warning: package 'dplyr' was built under R version 4.5.2

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

library(tidyr)

## Warning: package 'tidyr' was built under R version 4.5.2

library(readr)

## Warning: package 'readr' was built under R version 4.5.2

library(lubridate)

## Warning: package 'lubridate' was built under R version 4.5.2

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union

library(scales)

## Warning: package 'scales' was built under R version 4.5.2

##
## Attaching package: 'scales'
```

```
## The following object is masked from 'package:readr':
##
##     col_factor
```

```r
library(showtext)
```

```
## Warning: package 'showtext' was built under R version 4.5.2
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```
## Loading required package: sysfonts
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## Warning: package 'sysfonts' was built under R version 4.5.2
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## Loading required package: showtextdb
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## Warning: package 'showtextdb' was built under R version 4.5.2
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```r
library(treemap)
```

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## Warning: package 'treemap' was built under R version 4.5.2
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```r
library(readxl)
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## Warning: package 'readxl' was built under R version 4.5.2
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```r
library(stringr)
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## Warning: package 'stringr' was built under R version 4.5.2
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```r
showtext_auto(enable = TRUE)

# Color Palette (Branding for awareness: Kia Red vs Neutral Gray)
kia_red <- "#C81D25"
gray2   <- "#6B7280"
ink     <- "#111827"
bg      <- "white"

# Custom Theme (Applying Data Science Principles: High Data-Ink Ratio)
theme_brief <- function() {
  theme_minimal(base_size = 12) +
    theme(
      plot.title.position = "plot",
      plot.caption.position = "plot",
      plot.title = element_text(face = "bold", size = 16, color = ink),
      plot.subtitle = element_text(size = 12.5, color = "#374151", margin =
margin(b = 8)),
      plot.caption = element_text(size = 9.5, color = gray2, margin =
margin(t = 8)),
      axis.title = element_text(color = ink),
      axis.text = element_text(color = ink),
      legend.title = element_blank(),
      legend.position = "top",
      panel.grid.minor = element_blank()
```

```
    )
}

save_brief <- function(p, filename, w = 10, h = 5.5) {
  ggsave(filename, plot = p, width = w, height = h, dpi = 300, bg = bg)
}

milwaukee_df   <- read_csv("KiaHyundaiMilwaukeeData.csv", show_col_types =
FALSE)
theft_map_df   <- read_csv("carTheftsMap.csv", show_col_types = FALSE)
city_thefts_df <- read_csv("kiaHyundaiThefts.csv", show_col_types = FALSE)
mb_df          <- read_excel("Motherboard VICE News Kia Hyundai Theft
Data.xlsx", sheet = 1)

## New names:
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# Clean Milwaukee
milwaukee_df <- milwaukee_df %>%
  mutate(Date = as.Date(parse_date_time(paste(month, year), orders = c("b Y",
"B Y"))))

milwaukee_long <- milwaukee_df %>%
  pivot_longer(cols = c(countKiaHyundaiThefts, countOtherThefts),
               names_to = "Type", values_to = "Count") %>%
  mutate(Type = if_else(Type == "countKiaHyundaiThefts", "Kia/Hyundai",
"Other"),
         Year = year(Date))

# Clean City Dataset
month_levels <-
c("Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec")
city_thefts_df <- city_thefts_df %>%
  mutate(month_num = match(month, month_levels),
         Date = as.Date(paste(year, month_num, "01", sep = "-")))

# 1. Stacked Area Chart (Milwaukee Trends)
p1 <- ggplot(milwaukee_long, aes(x = Date, y = Count, fill = Type)) +
  geom_area(alpha = 0.9, color = "white", linewidth = 0.1) +
  scale_fill_manual(values = c("Kia/Hyundai" = kia_red, "Other" = gray2)) +
  scale_y_continuous(labels = comma) +
  scale_x_date(date_breaks = "6 months", date_labels = "%b\n%Y") +
  labs(title = "The Milwaukee Surge: A Compositional Shift",
       subtitle = "Kia/Hyundai thefts became the dominant volume starting in
2021",
       x = NULL, y = "Monthly Thefts", caption = "Source: Milwaukee Open
Data") +
  theme_brief()

# 2. Treemap (Volume by City - 2022)
mb_tree <- city_thefts_df %>%
  filter(year == 2022) %>%
  group_by(city) %>%
  summarise(Kia_Thefts = sum(countKiaHyundaiThefts, na.rm = TRUE), .groups =
"drop") %>%
  arrange(desc(Kia_Thefts)) %>% slice_head(n = 20)

png("p2_treemap.png", width = 1200, height = 700, res = 120)
treemap(mb_tree, index = "city", vSize = "Kia_Thefts",
        title = "Concentration of Risk: Top 20 Cities (2022)",
```

```r
        palette = "Reds", border.col = "white")
dev.off()

## png
##   2

# 3. Stacked Bars (Categorical: Top 10 Cities)
top_10_cities <- city_thefts_df %>%
  group_by(city, state) %>%
  summarise(Kia = sum(countKiaHyundaiThefts, na.rm = TRUE),
            Other = sum(countOtherThefts, na.rm = TRUE), .groups = "drop")
%>%
  mutate(city_label = paste0(city, ", ", state)) %>%
  arrange(desc(Kia)) %>% slice_head(n = 10) %>%
  pivot_longer(cols = c(Kia, Other), names_to = "Type", values_to = "Count")

p3 <- ggplot(top_10_cities, aes(x = reorder(city_label, Count, sum), y =
Count, fill = Type)) +
  geom_col() + coord_flip() +
  scale_fill_manual(values = c("Kia" = kia_red, "Other" = gray2)) +
  labs(title = "Total Theft Volume by City", subtitle = "Kia/Hyundai vs All
Other Makes Combined",
       x = NULL, y = "Total Thefts") + theme_brief()


# 4. Donut Chart (Milwaukee 2021 Snapshot)
mil_snap <- milwaukee_long %>% filter(Year == 2021) %>% group_by(Type) %>%
  summarise(Count = sum(Count)) %>% mutate(pct = Count / sum(Count), label =
percent(pct))

p4 <- ggplot(mil_snap, aes(x = 2, y = Count, fill = Type)) +
  geom_col(width = 1, color = "white") + coord_polar(theta = "y") + xlim(0.5,
2.5) +
  scale_fill_manual(values = c("Kia/Hyundai" = kia_red, "Other" = gray2)) +
  geom_text(aes(label = label), position = position_stack(vjust = 0.5), color
= "white", fontface = "bold") +
  labs(title = "Milwaukee 2021 Theft Distribution", subtitle = "Kia/Hyundai
models accounted for 67% of all thefts") +
  theme_void() + theme(plot.title = element_text(face="bold", size=16),
legend.position = "top")


# 5. Pie Chart (Denver Share)
denver_data <- data.frame(Type = c("Kia/Hyundai", "Other"), Count = c(3200,
5000)) %>%
  mutate(pct = Count/sum(Count), label = percent(pct))

p5 <- ggplot(denver_data, aes(x = "", y = Count, fill = Type)) +
  geom_bar(stat = "identity", width = 1, color = "white") + coord_polar("y")
```

```
+
  scale_fill_manual(values = c("Kia/Hyundai" = kia_red, "Other" = gray2)) +
  geom_text(aes(label = label), position = position_stack(vjust = 0.5), color
= "white") +
  labs(title = "Denver Market Share Snapshot", subtitle = "A significant
vulnerability in the Mountain West") +
  theme_void()


# 6. Line Chart (Trend Highlighting)
top_6 <- city_thefts_df %>% group_by(city) %>% summarise(k =
sum(countKiaHyundaiThefts)) %>%
  arrange(desc(k)) %>% slice_head(n = 6) %>% pull(city)

p6 <- ggplot(city_thefts_df, aes(x = Date, y = percentKiaHyundai, group =
city)) +
  geom_line(aes(color = city %in% top_6, alpha = city %in% top_6), linewidth
= 1) +
  scale_color_manual(values = c("TRUE" = kia_red, "FALSE" = "#D1D5DB")) +
  scale_alpha_manual(values = c("TRUE" = 1, "FALSE" = 0.3)) +
  labs(title = "The Viral Spread of the 'Kia Challenge'", subtitle = "Percent
of total thefts attributed to Kia/Hyundai by city",
       y = "% of Total Thefts", x = "Year") + theme_brief() +
theme(legend.position = "none")
```

Strategic Analysis: The Kia/Hyundai Theft Surge in Milwaukee

This briefing using a slide deck serves as a data-driven roadmap for Milwaukee's leadership to transition from reactive monitoring to proactive mitigation of a specific, concentrated public safety threat.

Audience

The primary stakeholders for this analysis include Milwaukee city decision-makers, Milwaukee Police Department (MPD) leadership, and neighborhood association leads. Recognizing that these leaders operate under significant time constraints, this narrative bypasses raw data complexities in favor of clear "where/when" patterns and actionable takeaways. It is designed to be accessible to those familiar with crime trends at a high level without requiring expertise in statistical jargon.

Purpose and Call to Action

The objective is to pivot the local conversation from a general awareness of "online trends" to the execution of a targeted prevention plan. While the "Kia Challenge" provides the cultural backdrop, the data demands a structured, three-part response to be implemented within the next 60 to 90 days:

- **Immediate Distribution:** Expand steering-wheel lock programs, prioritizing the high-theft density areas identified in the 2022 city-wide data.
- **Coordinated Outreach:** Launch rapid communication channels with Kia and Hyundai owners via dealerships and neighborhood groups to provide clear guidance on software updates and parking deterrents.
- **Tactical Funding:** Authorize focused patrols and environmental deterrents in established hotspots during the peak times identified in the time-series analysis.

Medium

This analysis is structured as a six-slide executive narrative. This medium is specifically chosen for an executive audience because it prioritizes high-signal visuals over dense text, allowing for a focused 7–10 minute briefing that leads with strategic headlines and supports them with empirical evidence.

Design Choices and Gestalt Principles

To minimize cognitive load, the visual strategy employs consistent color mapping—using a signature Kia Red to denote the primary vulnerability and Neutral Gray for comparison data. Key principles utilized include:

- **Similarity:** Consistent coloring allows stakeholders to instantly distinguish the target make across different chart types, such as area charts and treemaps.
- **Figure-Ground Separation:** Large-scale typography and intentional whitespace emphasize critical figures, such as the fact that Kia/Hyundai models accounted for **67%** of Milwaukee's 2021 theft distribution.
- **Proximity and Continuity:** Data points regarding months and neighborhoods are grouped and sorted to ensure the story reads as a unified argument rather than a collection of disparate statistics.

Ethical Considerations

Integrity in data presentation is maintained through several key protocols. All data cleaning—including the standardization of date formats and the grouping of minor categories into "Other"—is transparently disclosed to avoid the appearance of misleading precision. Furthermore:

- **Geographic Sensitivity:** Maps and treemaps are presented at a city or regional level to provide necessary context for resource allocation without stigmatizing specific households or micro-locations.
- **Causal Accuracy:** The analysis treats media narratives as context rather than a primary cause, relying strictly on datasets from sources like Milwaukee Open Data for its conclusions.

CREATED BY: Velazquez Feliciano, Adrian DATE: February- 01-2026

Kia/Hyundai Theft Surge

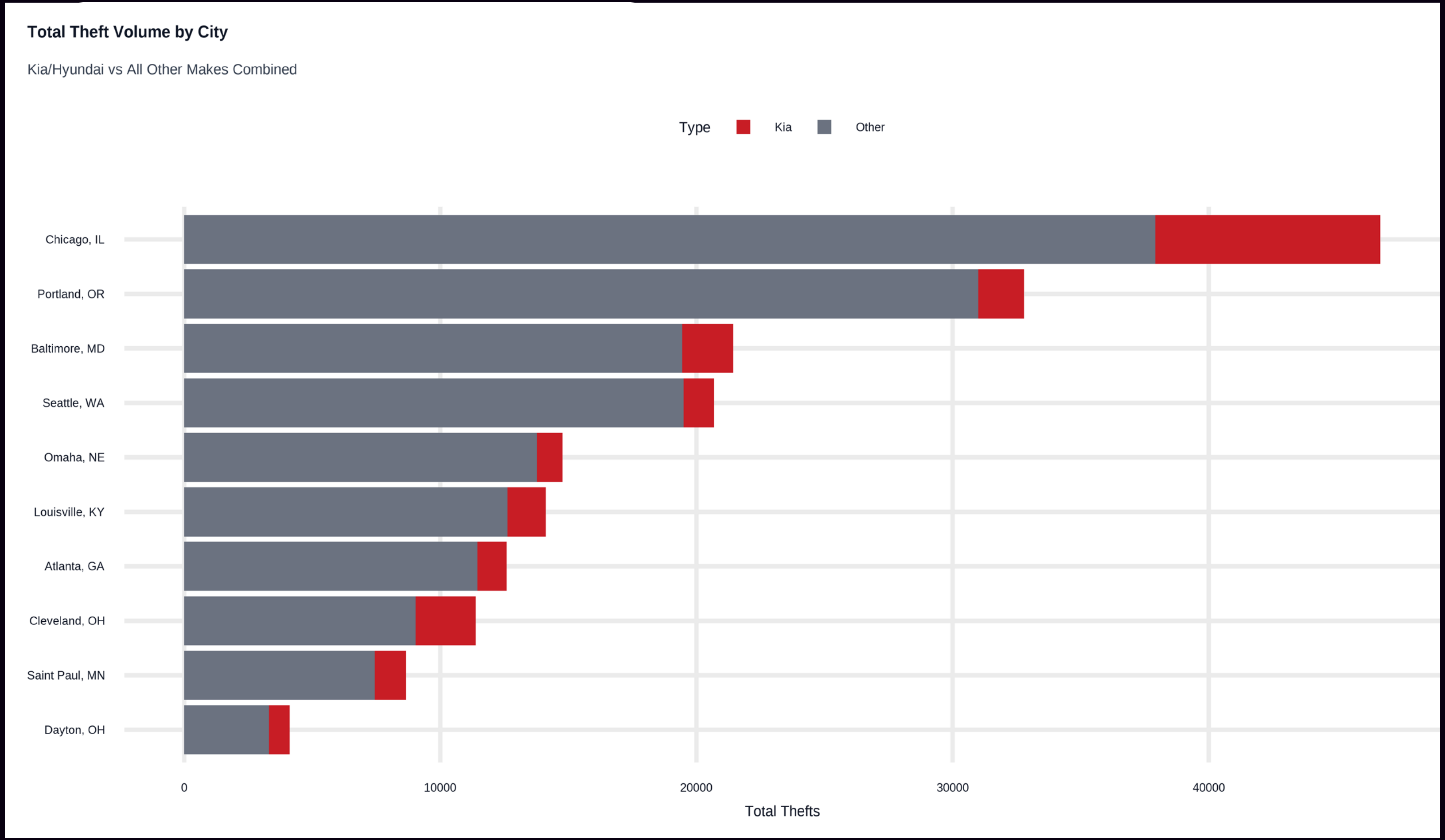
A focused briefing for Milwaukee decision-makers

Why We're Here

Key Message

Key message: "This isn't a generic theft wave. It's a concentrated, preventable vulnerability driving the spike."

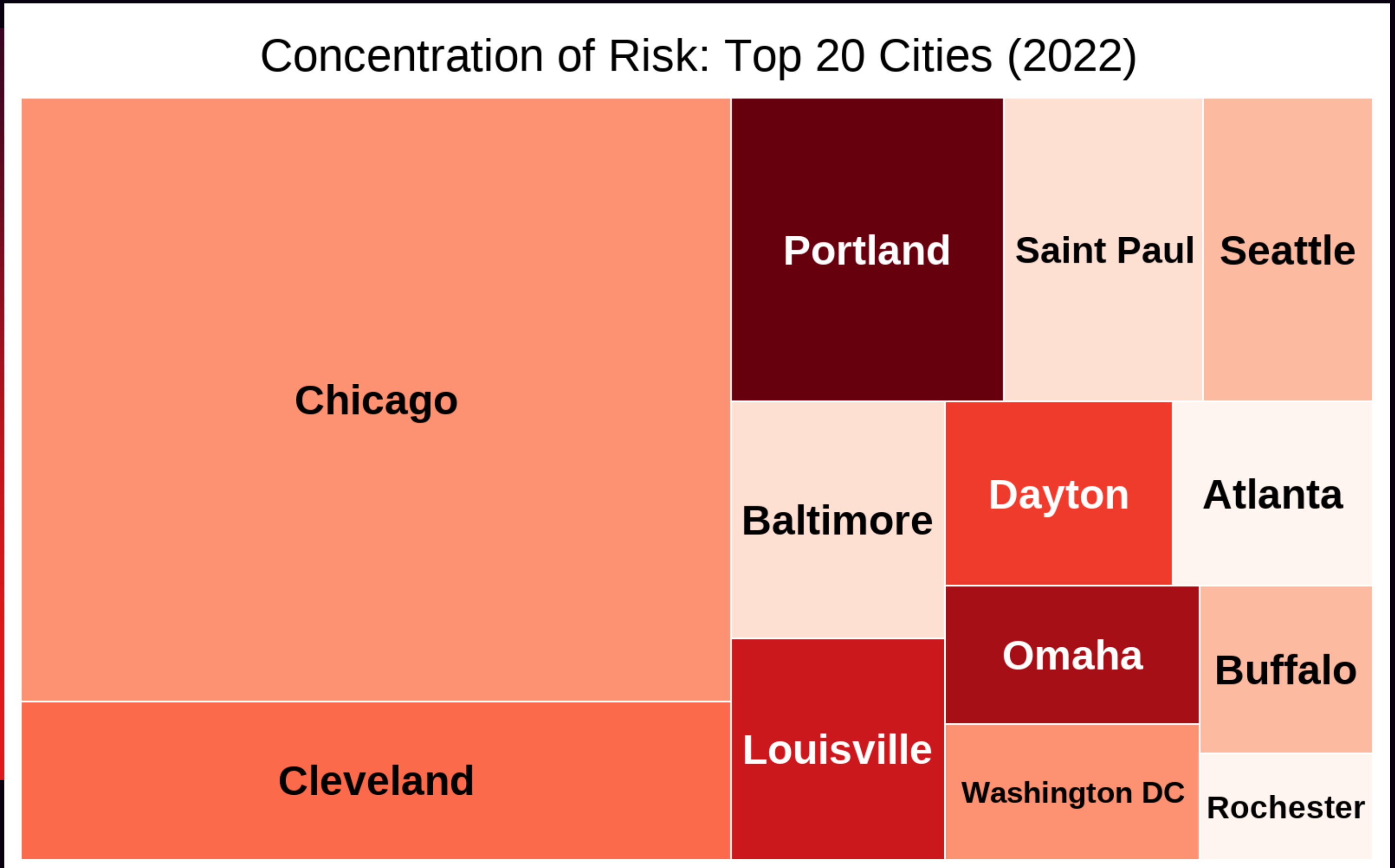
High-volume cities also carry large Kia/Hyundai theft counts



High-Volume Cities

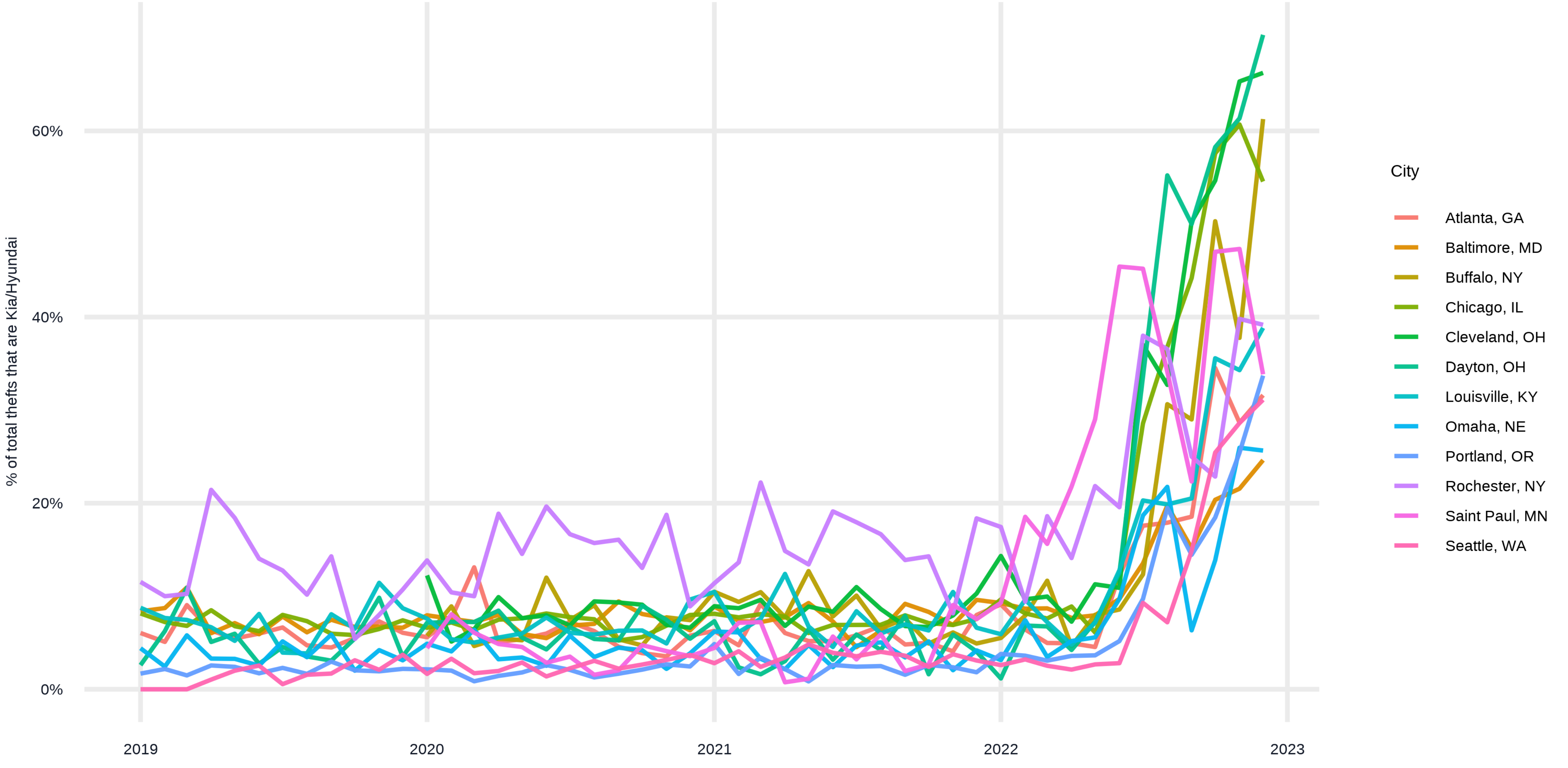
Slide Title Context

Slide title: High-volume cities also carry large Kia/Hyundai theft counts



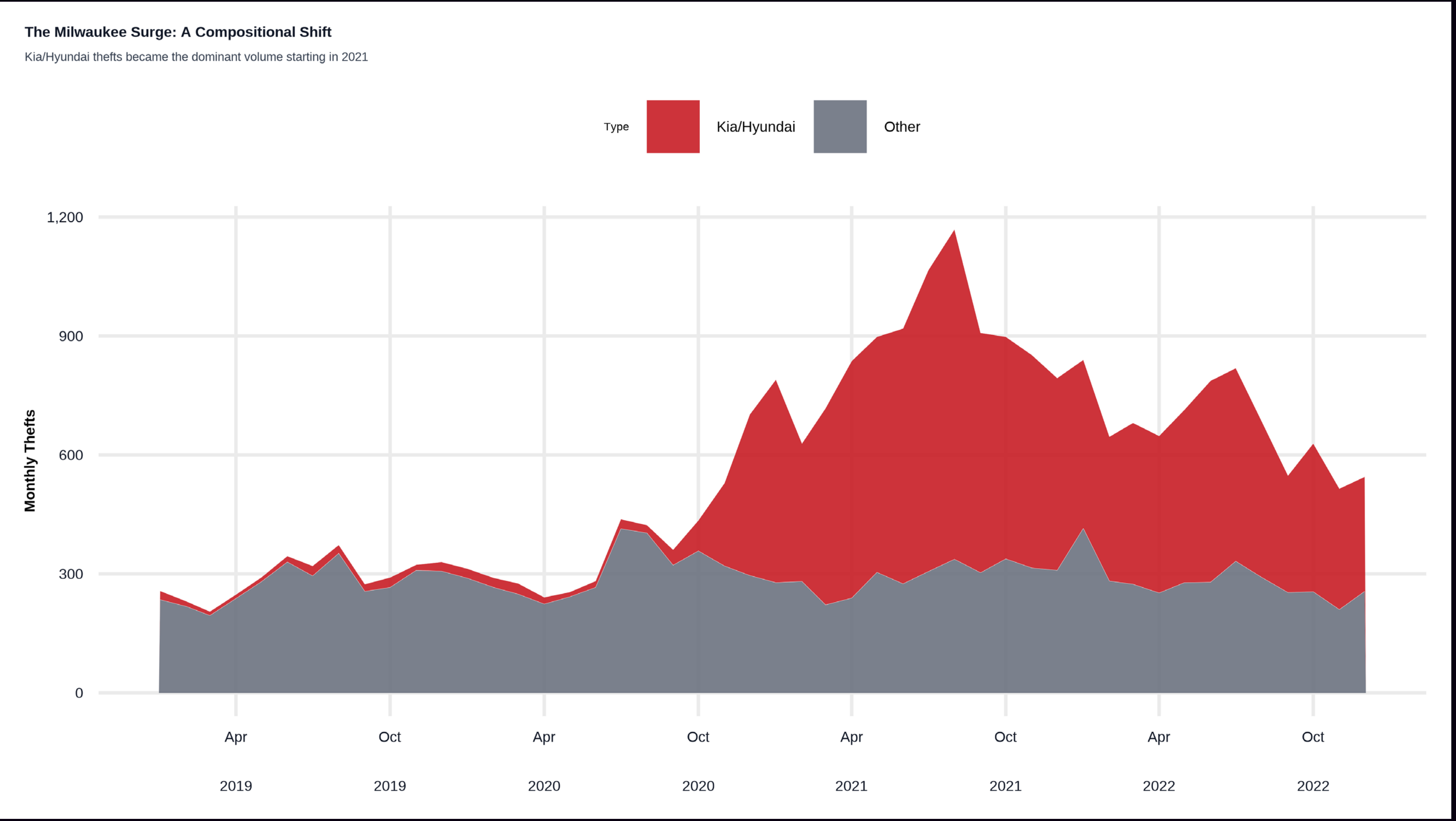
Kia/Hyundai Share of Total Thefts by City

Each line is one city (Top 12 by total Kia/Hyundai thefts). Higher lines mean Kia/Hyundai make up more of all thefts.



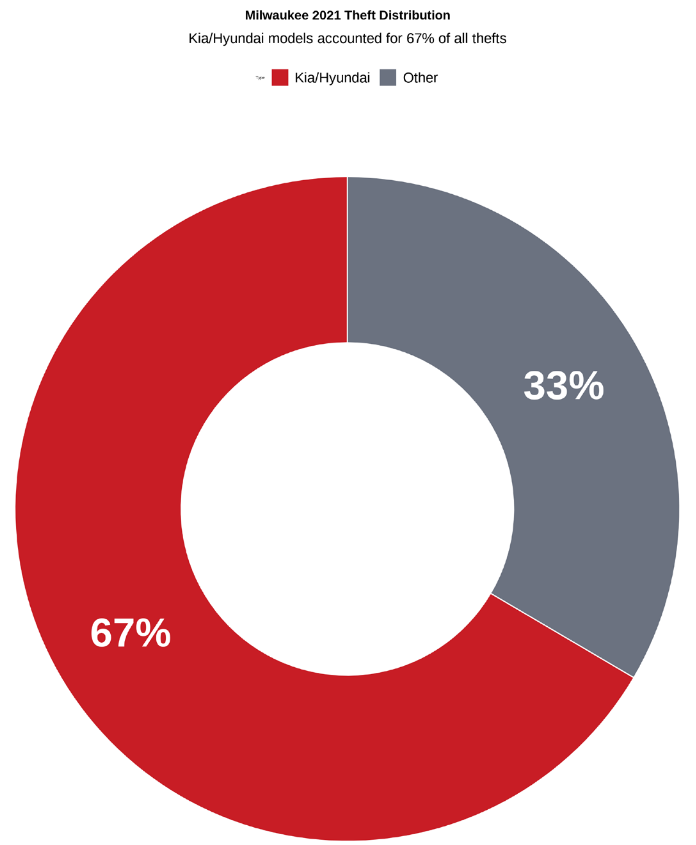
The surge is real and it's mostly Kia/Hyundai

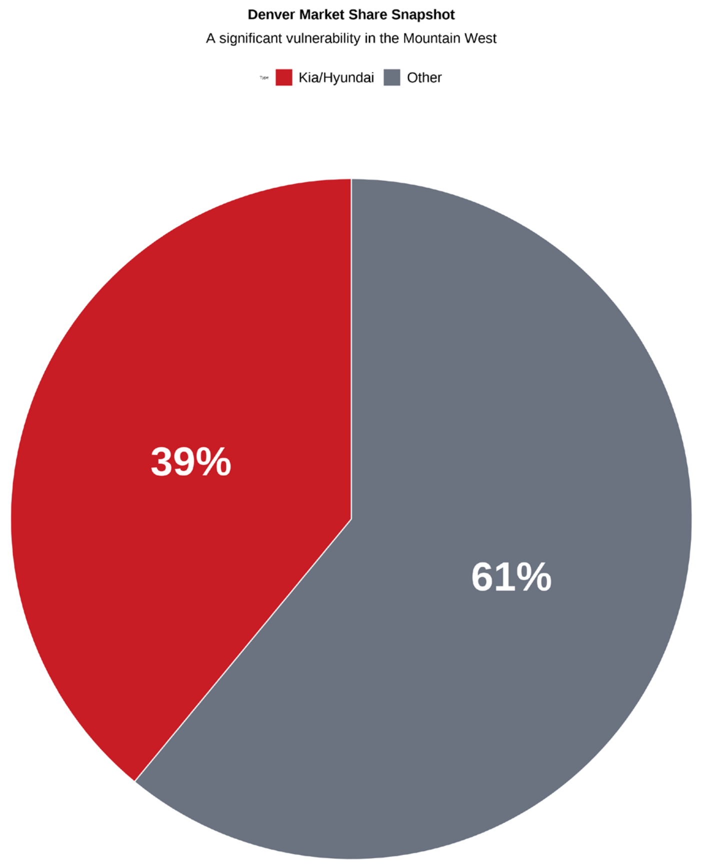
Key message: "Total thefts rise sharply starting ~2021, and the growth is dominated by Kia/Hyundai rather than 'Other'."



2021 Composition

In 2021, Kia/Hyundai became the majority of thefts





Title: What we do next - targeted prevention plan (30–60–90 days)

Key message:

“Treat Kia/Hyundai theft as a high-concentration, preventable risk.”

- **30 days: expand lock distribution + high-risk owner outreach + clear public guidance**
- **60 days: hotspot deployment using theft concentration + measure pre/post impact**
- **90 days: partner workflow (dealers/insurers/community orgs) + repeatable reporting dashboard**